GIGACORE

COLD PLASMA HYDROGEN PRODUCTION

BREAKTHROUGH TECHNOLOGY RESHAPING DECARBONIZATION

INSTITUTIONAL INVESTOR MEMORANDUM

Series A: **$20M** · **33% Pure Equity** · Post-Money: **$60.6M**

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01 EXECUTIVE SUMMARY & INVESTMENT DASHBOARD

Cold Plasma Reactor Technology — Green Hydrogen Production

CONVENTIONAL ELECTROLYSIS

50–55 kWh/kg

LCOH $4.40/kg
Alkaline · PEM · SOEC
Precious metal catalysts required

GIGACORE COLD PLASMA REACTOR

30–35 kWh/kg

Target LCOH $2.75/kg (pilot-stage)
Non-thermal plasma + metal reaction
No precious metals. Near-ambient operation

30%
Less energy than conventional electrolysis (observed)

38%
Lower LCOH target ($2.75 vs $4.40/kg)

99.99%
Hydrogen purity output

Zero
Precious metal catalysts required

30–35 kWh/kg
Observed Energy Range
Target 25 kWh/kg at pilot scale

$2.75/kg
LCOH Target
Parity with grey ammonia (pilot certification required)

TRL 5–6
Technology Readiness
Lab-validated · pre-commercial pilot stage

$20M
Series A Raise
Full Phase 1 — 18-month pilot programme

33%
Pure Equity, No Prefs
No ratchet, no liquidation priority

3–5×
Target Return Multiple
Valuation re-rating on pilot validation

INVESTMENT THESIS AT A GLANCE

METRIC	VALUE	SIGNIFICANCE
Current Performance	30–35 kWh/kg	Multiple successful runs confirmed
Target Performance	25 kWh/kg (commercial pilot)	Unlocks $2.25/kg LCOH subject to pilot validation
Current Status	TRL 5–6 Lab Validated	Bench-scale proven; structured pilot programme next
Market Inflection	$2.75/kg cost parity	Ammonia & refining parity relevant addressable segment
Series A Raise	$20M (full Phase 1)	18-month runway covering pilot, demo, team, BD

Investor Equity	33% pure equity	No preferences, no ratchet, no liquidation priority
Post-Money Valuation	$60.6M	Pre-pilot, pre-demo, pre-customer venture-stage entry point
Strategic Exit Target	36–60 months post-pilot validation	IP acquisition by Tier 1 industrial; multiple credible acquirers
Exit Framing	Science/IP asset acquisition	Exit thesis is strategic IP value not EBITDA or revenue multiples
Return Basis	Valuation re-rating at milestones	Each validation milestone (pilot → cost cert → offtake) re-rates the asset
Primary Market	India: ammonia & refining	IFFCO, KRIBHCO cost parity unlocks immediate switching
IP Status	Patent application in preparation	Technology managed as trade secret with NDA-gated disclosure pending filing

ILLUSTRATIVE RETURN SCENARIOS ($20M RAISE · 33% EQUITY)

SCENARIO	TIMING	MULTIPLE	EXIT VAL.	INVESTOR	MOIC	~IRR
CONSERVATIVE	36–60 months post-pilot	3.0×	3× post-money	**$52M proceeds**	3.0×	55%
BASE CASE	36–60 months post-pilot	4.0×	4× post-money	**$69M proceeds**	3.9×	74%
UPSIDE	36–60 months post-pilot	5.0×	5× post-money	**$87M proceeds**	4.9×	90%

WHY GIGACORE ENERGY NOW?

Different Physics

Cold plasma observed at 30 to 35 kWh/kg vs 45 to 55 kWh/kg for conventional electrolysis. This is not an optimisation of existing technology. It operates on a different physical mechanism.

$2.75/kg Cost Threshold

At this price, green hydrogen reaches parity with grey ammonia. For fertilizer, refining, and steel producers, the cost delta becomes bankable at plant level.

Industrial Customer Discussions

India's largest fertilizer cooperative has published a net-zero roadmap. Preliminary commercial discussions are underway. A formal LOI depends on independent pilot validation. No commitment has been made.

Policy Support

India SIGHT ($530M), the EU hydrogen strategy, and the US IRA create non-dilutive funding pathways. GigaCore is targeting $2 to $5M in grants alongside renewable PPA support.

Modular Design

Units scale from 1 to 5 kg/hr up to 25 to 50 kg/hr. This reduces capex at each stage and allows faster deployment than monolithic electrolyser plants.

18-Month Validation Path

The pilot runs across three milestones: commissioning at Month 6, independent cost certification at Month 12, and demo commissioning at Month 18. Each milestone achieved improves the commercial and investment position.

Entry Point

Post-money valuation is $60.6M at $20M raise for 33% equity. This is pre-pilot, pre-demo, and pre-customer. The full upside from each validation milestone remains ahead.

Exit Pathway

The intended exit is acquisition by a Tier 1 industrial buyer: EPC groups, industrial gas producers, or large-scale ammonia and refining operators. Each has a clear rationale for acquiring validated hydrogen IP. Acquisition is the appropriate exit at this stage.

02 THE HYDROGEN INFLECTION POINT

Understanding Green Hydrogen's Economic Barrier

Green hydrogen has been technically possible for decades. The problem is cost. Electrolysers today run at 45 to 55 kWh/kg, which translates to $3.80 to $6.00/kg LCOH even at favourable electricity prices. Grey hydrogen costs $1.50 to $2.50/kg. That gap has not closed fast enough.

Fertilizer, steel, and refining companies cannot absorb a 50 to 100% cost premium. At $2.75/kg, green hydrogen reaches parity with grey ammonia. Below that price, decarbonisation becomes cost-neutral or better for the buyer.

▸ Below $2.75/kg, green hydrogen becomes cost-neutral for industrial buyers. Above it, adoption stalls. The pilot objective is to verify whether the plasma pathway reaches that threshold at 1 kg/hr scale, with independently certified cost data.

THE $2.75/KG COST PARITY THESIS

APPLICATION	H2 REQ.	GREY H2	GREEN @$2.75	DELTA (PREMIUM)	SEGMENT VALUE
Ammonia (Fertilizer)	176 kg/t	$264–440/t	$484/t	+$24–220/t (5–50%)	**Very large**
Steel (Direct Reduction)	60 kg/t	$90–150/t	$165/t	+$15–75/t (10–80%)	**Very large**
Refining (Hydrotreating)	20–30 kg/t	$30–75/t	$55–83/t	+$8–53/t (10–70%)	**Very large**
Chemicals (Specialty)	50–100 kg/u	Varies	Competitive	App-dependent	**Large**

03 TECHNOLOGY PLATFORM: COLD PLASMA HYDROGEN

Alternative Physics Pathway Lab Validation Status & Pilot Objectives

Conventional electrolysis is constrained by electrochemical thermodynamics. Water splitting has a theoretical minimum of 237 kJ/mol. Real systems add 0.5 to 1.5V of overpotential, pushing energy consumption to 45 to 55 kWh/kg. Twenty years of incremental work has reduced this by 2 to 3% per decade.

GigaCore is investigating a different pathway: non-equilibrium plasma activation to reduce the activation energy for water dissociation. Lab results show 30 to 35 kWh/kg, consistent with a meaningful efficiency advantage. The Series A pilot verifies this at 1 kg/hr scale using defined metrology standards and kill points.

CORE TECHNOLOGY ARCHITECTURE

COMPONENT	GIGACORE	CONVENTIONAL	ADVANTAGE
Voltage Approach	Pulsed (6–15V, variable)	Constant (1.5–3V)	Higher current density at lower average potential
Operating Temperature	Near-ambient (25–60°C)	Elevated (40–80°C)	Reduces cooling costs; improves efficiency
Electrode Material	Aluminium + polymer matrix	Precious metals (Ir, Pt, Ni)	Eliminates supply chain risk; reduces cost volatility
Cell Configuration	Modular stacks (1–5 kg/hr units)	Monolithic (50+ MW cells)	Enables rapid scaling; reduces manufacturing complexity
Hydrogen Purity	99.95%+ on first pass	99.9%+ (extra purification)	Lower downstream purification cost
Energy Efficiency	30–35 kWh/kg observed; 25 kWh/kg target	45–55 kWh/kg typical	20–45% advantage depending on target achievement; pilot validation required

LAB VALIDATION STATUS (TRL 5–6) CONFIRMED DATA POINTS

- ✔ Observed energy range: 30–35 kWh/kg across multiple experimental runs (target: 25 kWh/kg at pilot scale)
- ✔ Experiments successfully completed in all planned directions at laboratory level
- ✔ Modular reactor design (1 kg/hr base module) engineered and tested at lab scale
- ✔ Multi-module 5–10 kg/hr pilot configuration scoped and engineered

METROLOGY & VERIFICATION FRAMEWORK

Measurement integrity is central to investor credibility. The pilot verification protocol covers the following.

- → Hydrogen output: calibrated mass flow meters + cross-validated via gas chromatography (purity and yield)
- → Energy input: measured at DC power supply level with true RMS instrumentation (not estimated from nameplate)
- → Calorimetric cross-checking: required at pilot stage to close the energy balance
- → Error margin reporting: all efficiency figures reported with ±5% measurement uncertainty band
- → Independent third-party validation: commissioned at Month 12 of pilot (pre-certification)

NEXT PHASE TRL 6–7 MILESTONES (SUBJECT TO KILL POINTS)

- → 1 kg/hr base module commissioning in India (Month 6); build costs covered within Phase 1 Use of Funds
- → Scale to 5–10 kg/hr multi-module pilot (Month 12–18); 2-year total timeline from project start
- → 1,000-hour continuous endurance testing primary kill point for commercialisation go/no-go
- → Independent cost certification; gas chromatography yield verification
- → Regulatory pathway completion (ASME/CE/ISO depending on jurisdiction; compliance budget allocated)

04 MARKET OPPORTUNITY & ADDRESSABLE MARKETS

The Relevant Opportunity: Cost-Delta Economics, Not Global TAM

▸ At TRL 5 to 6, global hydrogen TAM figures mean nothing. What matters is the segment where sub-$2.75/kg hydrogen creates a bankable cost delta for ammonia producers and refiners. That is the market GigaCore targets.

INDIA: PRIMARY ADDRESSABLE SEGMENT (AMMONIA + REFINING)

SEGMENT	2024	2030	2035	TAM @$2.75/kg	KEY PLAYERS
Fertilizer (Ammonia)	0.85	1.5–2.0	2.5–3.0	**$4.1–8.3B**	India fertilizer cooperatives & state enterprises
Refining (Hydrotreating)	0.75	1.2–1.5	1.8–2.2	**$2.9–6.0B**	India national & private oil refiners
Steel (Direct Reduction)	0.25	0.6–0.8	1.2–1.5	**$1.6–4.1B**	Indian integrated steel producers
Chemicals & Specialty	0.10	0.2–0.3	0.5–0.8	**$0.8–2.0B**	Specialty chemical producers
TOTAL	—	3.5–4.5	6–8 MMT	**$9.4–20.4B**	Multiple sectors

INDIA GOVERNMENT SUPPORT SIGHT PROGRAM

INITIATIVE	BUDGET	DURATION	INCENTIVE	GIGACORE ELIGIBILITY
SIGHT Production Incentive	Rs. 4,440 Cr ($530M)	2024–2030	Up to Rs. 40/kg of green H2 produced	HIGH Sub-$2.75/kg aligns with program
Green Hydrogen PLI Scheme	Rs. 5,000 Cr	2024–2028	4% of manufacturing capex + 4%/year for 5 yrs	HIGH Modular reactors qualify
Renewable Energy PPA Support	State-level	Ongoing	Power purchase at $35–50/MWh (solar/wind)	HIGH Critical to LCOH economics
MNRE Direct Grants	Rs. 1,000+ Cr	Ongoing	Non-dilutive funding for R&D, pilots	HIGH Pilot funding $3–5M anticipated

05 LEVELIZED COST OF HYDROGEN (LCOH) ANALYSIS

Modelled Cost Structure Pending Pilot Certification

$4.40/kg	<$2.75/kg	20–45%
Conventional Electrolyser LCOH	**Target LCOH at Pilot Validation**	**Efficiency Advantage (Observed)**
Industry baseline (PEM/Alkaline)	Pilot-certified cost goal	Subject to 1,000-hr pilot confirmation

GIGACORE PLASMA SYSTEM LCOH PROJECTION

COST COMPONENT	PILOT (1 kg/hr)	ASSUMPTION BASIS	SENSITIVITY NOTE
Electricity (30 kWh/kg @ $50/MWh)	$1.50/kg	Team-confirmed: India solar PPA $35–50/MWh	Largest single cost component; drives parity case
Module CAPEX Amortisation	$0.40/kg	Module cost ÷ 5yr assumed life ÷ annual output	Sensitivity to actual reactor life is material
Compression & Storage	$0.15/kg	Industry standard estimate	Unchanged at scale
Water Treatment	$0.12/kg	Industry standard estimate	Minimal sensitivity
Maintenance & Consumables	$0.25/kg	Lab-derived; to be refined at pilot	Catalyst replacement frequency uncertain
Fixed Overhead	$0.20/kg	Early-stage estimate	Declines significantly at multi-module scale
Plasma Catalyst & Replacement	$0.15/kg	Included in maintenance above (split shown)	Subject to pilot durability data
TOTAL LCOH (30 kWh/kg observed)	**$2.77/kg**	**Pilot-stage model; not yet certified**	**At 25 kWh/kg target: ~$2.45/kg**

* GigaCore: 30–35 kWh/kg observed at lab scale; 25 kWh/kg is the pilot programme target. Not yet independently verified.

ELECTRICITY COST SENSITIVITY ANALYSIS

ELECTRICITY PRICE	3¢/kWh	5¢/kWh	7¢/kWh	10¢/kWh
Electricity Cost (30 kWh/kg observed)	$0.90/kg	$1.50/kg	$2.10/kg	$3.00/kg
Non-Electric Costs (pilot estimate)	$0.55/kg	$0.55/kg	$0.55/kg	$0.55/kg
TOTAL LCOH (30 kWh/kg basis)	**~$2.05/kg***	**$2.05/kg**	**$2.65/kg**	**$3.55/kg**

At 25 kWh/kg target (if achieved)	~$2.00/kg*	~$2.00/kg*	$2.29/kg	$3.00/kg
Meets <$2.75/kg without subsidy?	Yes	Yes (if 25 kWh/kg)	Yes (30 kWh/kg)	No subsidy required

† Figures marked with † are contingent on achieving 25 kWh/kg at pilot scale and sustaining 1,000-hour continuous operation. These are NOT certified cost claims and should not be represented as such to third parties.

▸ All LCOH figures are modelled estimates based on 30 to 35 kWh/kg observed in lab conditions. These are not certified cost claims. Third-party certification is a pilot milestone. Treat these as illustrative inputs, not validated outputs.

PRELIMINARY CAPEX ASSUMPTIONS (SUBJECT TO PILOT VALIDATION)

LCOH figures are only credible when capex assumptions are stated clearly. The following inputs are confirmed by the technical team and used in the model.

ITEM	ASSUMPTION	NOTE / STATUS
1 kg/hr base module build cost	**Detailed in Use of Funds (Section 13)**	1-year build timeline; costs fully budgeted in Series A
5–10 kg/hr multi-module pilot	**Detailed in Use of Funds (Section 13)**	1-year build timeline; covered within Phase 1 budget
Electricity cost (India)	**$0.035–$0.050/kWh**	Solar/wind PPA rates; state-level agreements
Estimated reactor service life	**Minimum 5 years**	Assumed for modelling; to be validated at pilot stage
Depreciation basis	**10 years straight-line**	Standard industrial assumption; capex-sensitive to actual life
Module CapEx per kW	**$150–$250/kW**	Estimated; to be refined post-pilot build
Annual maintenance & consumables	**$0.15–0.25/kg H2**	Includes catalyst replacement; based on lab data

Reactor durability beyond 5 years has not been demonstrated. The model uses 5 years as the conservative baseline. If actual service life reaches 8 to 10 years, LCOH improves materially. Apply appropriate uncertainty to all capex-dependent outputs until the pilot provides data.

06 COMPETITIVE LANDSCAPE & DIFFERENTIATION

Why Conventional Electrolysers Cannot Reach $2.75/kg

TECHNOLOGY	kWh/kg	$/kW	TRL	PREC. METALS	<$2.75?	KEY LIMITATION
Alkaline Electrolysis	50–55	$400–800	9	None	NO	Efficiency plateau electrochemistry limits
PEM Electrolysis	45–55	$700–1200	8	Iridium (15–20%)	NO	Precious metal constraint; durability issues
AEM Electrolysis	45–55	$600–1000	7	Nickel/Cobalt	NO	Same efficiency limit; capped at ~15MW/stack
SOEC (High-Temp)	40–48	$800–1500	5–6	Nickel/LSM	UNLIKELY	Capital intensity; thermal complexity
GigaCore Plasma ✓	**25–35 ***	$150–250	5–6	NONE ✓	**YES ✓**	30–35 kWh/kg observed; 25 kWh/kg pilot target

PHYSICS BARRIER TO INCREMENTAL IMPROVEMENT

The core problem is thermodynamic. Water splitting requires at least 237 kJ/mol. Conventional electrolysis adds 0.5 to 1.5V overpotential on top of that, pushing real consumption to 45 to 55 kWh/kg. Twenty years of research has improved this by 2 to 3% per decade.

Plasma activation works differently. It uses ionisation-induced fragmentation rather than electrochemical kinetics. If the pilot confirms this at scale, it represents a structural efficiency advantage over electrolyser-based pathways.

07 TECHNOLOGY READINESS & VALIDATION ROADMAP

De-Risking Path: TRL 5 to Commercial (TRL 8–9)

TRL	DELIVERABLE / MILESTONE	TIMELINE	CAPITAL	KEY DE-RISKING METRIC	VALUATION INFLECTION
TRL 5–6	**Pilot reactor (5–25 kg/day) operational in India**	Months 0–6	$7–8M	30 kWh/kg sustained; 500+ hrs durability	**Baseline**
TRL 6–7	**100 kg/day demo plant; cost certification completed**	Months 6–18	$12–13M	Independent cost & efficiency validation; commercial discussions formalised	**Potential valuation re-rating on third-party validation**
TRL 7–8	**Commercial module replicated; second site engineering**	Months 18–24	Series B tranche: $15–20M	Industrial JV agreement (if validation successful); second customer discussions	**Potential re-rating on binding offtake + cost certification**
TRL 8–9	**Multi-site commercial operation (1,000+ kg/day aggregate)**	Months 24–36	Series B: $50–75M	Three-site; $50M+ revenue; EBITDA breakeven path	**Series B positioning improves materially on commercial milestones**

DEFINED KILL POINTS GO / NO-GO CRITERIA

The pilot has pre-defined binary kill points. If any criterion fails, next-phase investment halts pending technical review.

KILL POINT CRITERION	THRESHOLD	CONSEQUENCE IF FAILED
Sustained energy efficiency at pilot scale	>32 kWh/kg over 1,000 continuous hours	Technology non-competitive vs. AEM at <$400/kW; programme suspended
Performance degradation over duty cycle	>10% drop over any 500-hour cycle	Catalyst replacement economics break down; commercialisation halted
CapEx vs. modelled envelope	>30% above budgeted module cost (per Use of Funds)	LCOH model breaks; requires engineering redesign before Series B
Hydrogen yield measurement integrity	Variance >±8% vs. GC cross-validation	Metrology protocol failure; independent audit required before proceeding
Regulatory certification pathway	No viable path within 18 months	Jurisdictional pivot required; India deployment delayed

08 COMMERCIALISATION STRATEGY

3-Phase Rollout Pilot to Commercial

REVENUE & BUSINESS MODEL

The primary business model after validation is JV manufacturing with an anchor industrial partner: co-build and operate a hydrogen plant with an offtake commitment. Technology licensing to EPC firms is the secondary path. Equipment OEM is possible but not the near-term focus.

MODEL	DESCRIPTION	WHEN / TRIGGER
Equipment Sale	Sale of 1 kg/hr and 5–10 kg/hr plasma hydrogen modules to industrial customers	Post-pilot validation (Year 2+); pricing TBD at pilot stage
JV Manufacturing	Joint venture with anchor industrial partner to co-build and operate hydrogen plant	JV timeline contingent on pilot validation outcomes no committed date
Technology Licensing	License plasma reactor IP and operating method to EPC partners and regional manufacturers	Post-Series B; jurisdictions: India, EU, GCC, North America
IP Asset Value	Technology managed as trade secret with NDA-gated disclosure	Patent application in preparation; filing pre-pilot commissioning

PHASE 1: PILOT VALIDATION & FOUNDATION (MONTHS 0–18) FULL SERIES A DEPLOYMENT

Objective: operate a 100 kg/day hydrogen pilot, validate cost economics, secure non-dilutive grants, and complete third-party cost certification. The full $20M Series A is deployed across 18 months covering pilot construction, R&D, team, business development, and working capital.

3 **Pilot reactor fabrication begins; EPC partner onboarded**

6 **Pilot commissioning in Pune; first hydrogen production**

9 **Sustained 100 kg/day operation; cost per kg measured**

1 **MNRE grant application submitted; SIGHT PLI pre-approval**

1 **Third-party cost certification; independent validation report published**

PHASE 2: DEMONSTRATION & JV INCEPTION (MONTHS 18–30)

Objective: a 500 kg/day demonstration plant and progress toward a formal JV framework with an industrial partner. Estimated capital of $6 to $8M from a Series B tranche or non-dilutive grants.

→ Month 18 to 20: industrial JV framework discussed. Formal term sheet depends on pilot data

→ Month 20–24: EPC partner selected; site prepared; equipment procured

- → Month 24–28: Demo plant commissioned; hydrogen delivery to IFFCO ammonia synthesis begins
- → Month 28 to 30: replication feasibility study for a second site

PHASE 3: COMMERCIAL ROLLOUT & SERIES B (MONTHS 30–36+)

Objective: multi-site commercial deployment across 3 to 5 plants. Series B of $50 to $75M (indicative) to fund expansion.

- → Month 30 to 32: second JV term sheet signed with a regional industrial customer
- → Month 32–36: Series B close; strategic investor + growth equity participation
- → Month 36+: Three-site operation; 1,000+ kg/day aggregate hydrogen production
- → Month 36+: Strategic exit process initiated M&A with Tier 1 industrial gas, energy, or EPC acquirer

09 INDIA MARKET: ANCHOR CUSTOMER & EXPANSION

IFFCO Indian Farmers Fertiliser Cooperative

1.9M t/yr	3–5¢/kWh	$530M
India H2 Demand	**Renewable Power Cost**	**India SIGHT Program**
Fertilizer + refining alone	India solar/wind (vs 6–10¢ globally)	Government demand acceleration

ANCHOR CUSTOMER: IFFCO PROFILE

METRIC	VALUE	RELEVANCE TO GIGACORE
Annual Ammonia Production	900K tonnes/year	Largest ammonia producer in India; 176 kg H2/tonne = 158.4K tonnes H2/year demand
Current Hydrogen Source	100% grey hydrogen (gas)	Requires decarbonisation; cost structure sensitive to green H2 pricing
Net-Zero Target	2030 roadmap published	Corporate mandate: 30% green hydrogen blend by 2030
Capex Budget (Green H2)	Significant capex allocated (per published roadmap)	Financial commitment to hydrogen infrastructure; projects in planning stage
Offtake Potential	50–100 kg/day → 200+	First contract: Phase 2 demo (500 kg/day); expansion post-validation

CURRENT IFFCO ENGAGEMENT STATUS

- ✔ CEO-level introduction completed between GigaCore founder and senior IFFCO leadership. Preliminary stage only
- ✔ LOI discussions have been initiated. A formal LOI depends on independent pilot validation. Nothing has been signed
- ✔ Commercial framework discussions are ongoing. No terms have been agreed. All discussions are contingent on pilot outcomes
- ✔ JV structure is under discussion. A formal agreement is only possible after pilot validation and cost certification
- ✔ First hydrogen offtake depends on a formal JV agreement and successful demo commissioning. No timeline is committed

10 VALUATION FRAMEWORK & EXIT SCENARIOS

Series A Pure Equity Structure

EXIT SCENARIO ANALYSIS (36–60 MONTHS POST-PILOT VALIDATION)

SCENARIO	TRIGGER CONDITIONS	TARGET MULTIPLE	ILLUSTRATIVE PROCEEDS
CONSERVATIVE	Pilot validated at 30–35 kWh/kg; cost independently certified; commercial discussions advanced	3× post-money	~$52M (illustrative contingent on exit timing and terms)
BASE CASE	Industrial JV agreement reached; 1,000-hr demo validated; binding offtake signed	4× post-money	~$69M (illustrative contingent on exit timing and terms)
UPSIDE	Two industrial JV agreements; Tier 1 strategic acquirer engaged; Series B at uplift	5× post-money	~$87M (illustrative contingent on exit timing and terms)

IRR figures assume a 4 to 5 year hold. Exit timing in heavy industrial hydrogen cannot be reliably projected at TRL 5 to 6. Institutional buyers will need 12 to 24 months of post-demonstration data, independent cost certification, and sustained 1,000+ hour durability before any acquisition.

STRATEGIC ACQUIRER ANALYSIS

ACQUIRER	EXAMPLES	STRATEGIC RATIONALE	MULTIPLES	GIGACORE FIT
Major EPC / Engineering	Tier 1 EPC & engineering groups	Add hydrogen tech to project delivery and licensing portfolio	8–12× revenue	Strong strategic fit modular design aligns with EPC replication model
Diversified Energy	International oil & gas majors	Decarbonisation portfolio; hydrogen cost reduction mandate	5–10× revenue	Reasonable fit complements industrial hydrogen cost reduction mandates
Industrial Gas Majors	Industrial gas producers	Cost curve leadership in green hydrogen production	6–12× revenue	Strong strategic fit IP acquisition complements electrolyser market position
Industrial End-Users	Large-scale ammonia & refining producers	Backward integration; captive sub-$2.75/kg H2 supply	4–8× revenue	Logical acquirer profile backward integration rationale; first-mover advantage post-JV

11 INVESTOR RETURNS & FINANCIAL PROJECTIONS

Illustrative Model Exit Thesis is IP Acquisition, Not EBITDA

This financial model is illustrative. The exit thesis is IP acquisition by a Tier 1 industrial buyer, not EBITDA growth. Revenue projections are included for context. Investors should not base return expectations on these numbers.

METRIC	YEAR 1 (Pilot)	YEAR 2 (Demo)	YEAR 3 (Scale)	YEAR 4 (Multi-site)	YEAR 5 (Expansion)
Installed Capacity (kg/day)	100	500	2,000	3,500	5,500
Utilisation Rate	60%	70%	80%	82%	85%
Annual H2 Production (t)	20	130	620	1,050	1,700
H2 Selling Price ($/kg)	—	$4.50	$4.20	$3.95	$3.70
Revenue ($M)	$0	**$0.59**	**$2.60**	**$4.15**	**$6.29**
Total COGS ($/kg illustrative)	TBD	$2.20	$2.10	$2.05	$2.00
Gross Profit ($M)	$0	$0.12	$0.80	$1.21	$1.97
Gross Margin	N/A	20%	31%	29%	31%
Operating Expenses ($M)	$2.5	$2.8	$3.2	$3.8	$4.5
EBITDA ($M)	-$2.50	-$2.21	-$2.40	-$2.59	-$2.53
Capex ($M)	$5.0	$4.5	$3.0	$2.5	$2.0

12 RISK ANALYSIS & MITIGATION STRATEGIES

Technical, Commercial & Financial Risks

TECHNICAL RISKS

RISK	IMPACT	MITIGATION
Reactor durability insufficient failure to sustain performance at industrial duty cycles would eliminate economic advantage and render technology non-competitive	LCOH +$0.15–0.30/kg; programme suspended at kill point	Monthly durability testing; 1,000-hour kill point defined; dedicated R&D team
Pilot cost overruns (15–20%)	Schedule slip 3–6 months; budget pressure	Fixed-price EPC contracts; stage-gate funding; $2.5M contingency in budget
Pilot efficiency underperforms target (30 vs 25 kWh/kg)	LCOH higher than modelled; still potentially sub-$2.75 at 30 kWh/kg at low electricity cost	Conservative base case uses 30 kWh/kg observed; 25 kWh/kg is upside only
Regulatory certification delays (ASME/CE/ISO)	Commercialisation delayed 3–6 months	Early engagement with notified bodies; compliance budget allocated

COMMERCIAL & FINANCIAL RISKS

RISK	IMPACT	MITIGATION
Anchor customer negotiations fail or delay	Pivot to alternate industrial buyer; adds 6–12 months	Preliminary discussions with multiple India fertilizer and refining companies; no single-customer dependency
AEM electrolyser reaches 40–42 kWh/kg at <$400/kW (projected; AEM currently ~55 kWh/kg)	Plasma structural advantage narrows; kill point triggered	Kill point: plasma must sustain <30 kWh/kg to retain advantage; defined go/no-go
Green H2 subsidies cut (policy risk)	Customer threshold rises; delays adoption	Target <$2.75/kg without subsidy at pilot scale not subsidy-dependent by design
Key personnel departure (CTO / lead scientists)	Execution slowdown; IP integrity risk	14-person core team; technical IP company-owned; CVs available under NDA
FX risk (INR/EUR/USD volatility)	India operations cost change 5–10%	Hedging via forward contracts; capex front-loaded (Months 0–6)

MEASUREMENT INTEGRITY RISK

Institutional investors will scrutinise whether efficiency claims reflect real physics or measurement artefacts. GigaCore addresses this directly.

RISK VECTOR	MANAGEMENT APPROACH
Gas mixing artefacts	Cross-validate H2 yield via gas chromatography at every measurement interval; not mass-flow alone
Overstated hydrogen yield	Calorimetric energy balance closure required at pilot stage yield and energy input must reconcile
Electrical measurement error	True RMS instrumentation at DC supply level; not nameplate or estimated values; ±5% uncertainty band reported

Patent application is being prepared ahead of pilot commissioning. Until filing is confirmed, the technology is protected as trade secret with NDA-gated disclosure. Investors will receive filing date, jurisdictions, and claims scope as part of Series A documentation.

13 USE OF FUNDS & DEPLOYMENT SCHEDULE

$20M Raise Full Phase 1 Deployment (18 Months)

CATEGORY	M 0–6	M 6–12	M 12–18	TOTAL ($M)	% RAISE
PILOT REACTOR CONSTRUCTION & COMMISSIONING	$2.0M	$3.0M	$1.5M	**$6.5M**	32.5%
— Plasma reactor modules & control systems	$1.0M	$1.8M	$0.7M	$3.5M	17.5%
— Site civil, electrical, safety infrastructure	$0.6M	$0.8M	$0.5M	$1.9M	9.5%
— H2 compression, storage, safety certification	$0.4M	$0.4M	$0.3M	$1.1M	5.5%
R&D & TECHNOLOGY VALIDATION	$0.6M	$0.8M	$0.6M	**$2.0M**	10.0%
TEAM EXPANSION (Germany + India)	$0.8M	$1.0M	$0.9M	**$2.7M**	13.5%
BUSINESS DEVELOPMENT & CUSTOMER ACQUISITION	$0.4M	$0.6M	$0.5M	**$1.5M**	7.5%
OFFICE, PERMITS & COMPLIANCE	$0.3M	$0.4M	$0.3M	**$1.0M**	5.0%
CONTINGENCY & WORKING CAPITAL	$1.0M	$0.8M	$0.7M	**$2.5M**	12.5%
TOTAL DEPLOYMENT (18 MONTHS)	$5.1M	$6.6M	$4.5M	**$16.2M**	81.0%
Strategic Reserve / Series B Bridge	—	—	—	$3.8M	19.0%
TOTAL RAISE	—	—	—	**$20.0M**	100%

14 MANAGEMENT & GOVERNANCE

Leadership Team & Governance Structure

LEADERSHIP TEAM

NAME	ROLE	BACKGROUND
Manish Kothary	**Chief Executive Officer**	Founder developing scalable hydrogen and plasma-based energy systems for industrial decarbonization. Focused on high-efficiency reactor platforms, modular deployment and strategic partnerships to enable disciplined commercial scale-up. Previously founded Bene LLC; earlier senior roles at Barclays Global Investors, Wells Fargo and Pacific Gas & Electric. M.Sc Computer Science, University of South Carolina; BE Electrical Engineering, IIT Roorkee.
Vandit Verma	**President Infrastructure & Energy Platforms**	Leads global infrastructure development and commercialisation of advanced energy platforms. Cross-border project structuring, strategic capital partnerships. Experience in hydrogen and low-carbon fuel systems, rare earth extraction and waste-to-energy infrastructure across North America, Europe and emerging markets. B.Eng Mechanical Engineering, Ryerson University, Canada.
Igor Kuntevitsky	**Chief Technology Officer**	Leads reactor architecture, hydrogen systems engineering and IP development. Inventor of multiple granted U.S. patents covering hydrogen production and critical mineral recovery systems (hydrocarbon gas pathways, red mud processing). Expertise spans matrix reactor systems, controlled partial oxidation and integrated hydrogen purification.
Prashanth Shetty	**Business Development**	Leads business development across Gigacore's hydrogen and energy infrastructure platform, driving institutional investor engagement, commercialization strategy and strategic partnerships. Former executive leader at Verdigris, EMC and SAP with experience spanning global go-to-market strategy and energy technology platforms. MBA, The Wharton School; MS Engineering, Stanford University.
Ajeet Saran	**EPC Director & Government Relations India**	Leads project execution, regulatory interface and industrial deployment in India. Industry leader with extensive experience across coal, steel, logistics and EPC project execution. Spent nearly 25 years at Tata Steel across mining, power, sales and supply chain, bringing deep techno-commercial expertise and strong government liaison capability. BE Electrical Engineering, IIT-BHU; PG Supply Chain, XLRI.
Gautam Ghosh	**Director Finance, Tax & Compliance**	Oversight on financial governance, cross-border structuring and regulatory compliance. Former Head of Taxation (South West Asia & Middle East), Motorola Solutions India. Prior senior roles at Coca-Cola India and Nestlé India. Career began at Price Waterhouse advising MNCs on India entry strategy. Chartered Accountant; B.Com (Hons), SRCC, Delhi University.

The core technical team is 14 full-time engineers and scientists. Their backgrounds cover plasma physics, materials science, nanostructured media, and industrial reactor engineering. CVs are available under NDA.

The CTO, Igor Kuntevitsky, holds multiple granted U.S. patents in hydrogen production systems. GigaCore's cold plasma method and reactor architecture are subject to a patent application currently in preparation. In the interim, the technology is protected as a trade secret with NDA-gated disclosure.

GOVERNANCE & INVESTOR RIGHTS

GOVERNANCE ITEM	STRUCTURE
Board Composition	3 seats: Founder (1), Investor (Observer), Independent (TBD)
Board Meetings	Monthly during 18-month Series A deployment; then quarterly
Investor Rights	Information rights (monthly P&L, quarterly board packs); pro-rata rights in Series B
Anti-Dilution Protection	Broad-based weighted average standard (typical for Series B; not requested for Series A)
Liquidation Preference	NONE pure equity structure; all shareholders treated equally on exit
Drag-Along / Tag-Along	Standard terms for M&A exit
Reporting & Transparency	Monthly P&L to investors; quarterly board materials; annual audit

15 INVESTMENT HIGHLIGHTS & DECISION MATRIX

Why GigaCore, Why Now, Why This Structure

01 TECHNOLOGY PROGRAMME

Cold plasma observed at 30–35 kWh/kg vs 45–55 kWh/kg for PEM/alkaline electrolysers. Represents a potential 20–35% efficiency improvement subject to pilot validation.

02 MARKET INFLECTION POINT

$2.75/kg hydrogen crosses economic parity with grey ammonia. Fertilizer, steel, and refining decarbonisation becomes margin-neutral or value-accretive at this price. The investment thesis rests on cost-delta economics at plant level bankable without macro market assumptions.

03 INDUSTRIAL CUSTOMER DISCUSSIONS

India's largest fertilizer cooperative has published a net-zero roadmap. Preliminary commercial discussions have been initiated. Formal engagement contingent on pilot validation outcomes no LOI or JV committed.

04 18-MONTH VALIDATION PATH

Structured de-risking: pilot (Month 6) → independent cost certification (Month 12) → demo commissioning (Month 18). Each milestone achieved strengthens the commercial and investment case. No guaranteed outcome.

05 GOVERNMENT TAILWINDS

India SIGHT ($530M), EU hydrogen strategy, US IRA provide non-dilutive funding and demand certainty. Anticipated $2–5M in grants + renewable PPA support.

06 FAVOURABLE ENTRY POINT

$60.6M post-money ($20M raise · 33% equity) pre-pilot, pre-demo, pre-customer. Investors enter at maximum upside stage with downside controlled by kill-point structure.

07 CLEAR EXIT PATHWAY

Strategic M&A by a Tier 1 industrial EPC group, industrial gas producer, or large-scale end-user at post-pilot stage. Acquisition multiples for proven, cost-advantaged hydrogen IP typically 6–12× revenue. Series B at 3–5× post-money provides an intermediate liquidity event.

INVESTOR DECISION MATRIX: GIGACORE VS. ALTERNATIVES

FACTOR	GIGACORE (Plasma)	PEM Electrolysis	Alkaline Upgrade	SOEC (High-Temp)
Efficiency Potential (kWh/kg)	**25–30 ✓**	45–55 (plateau)	45–55 (plateau)	40–48 (uncertain)
Path to <$2.75/kg LCOH	**YES 24–30 mths ✓**	NO (5–7 years+)	NO (never)	UNLIKELY (20+ yrs)

TRL Today	**5–6 (Validated)**	8–9 (Mature)	9 (Mature)	5–6 (Early)
Industrial Customer Discussions	**Preliminary (pre-LOI)**	Multiple buyers	Multiple buyers	None identified
Indicative Return Multiple	**3–5× (illustrative)**	1.5–2×	1.2–1.5×	2–3×

▸ GigaCore is a structured verification programme at TRL 5 to 6. The case for investment rests on lab-observed efficiency of 30 to 35 kWh/kg, preliminary discussions with large industrial customers in India, a fully-funded 18-month pilot with defined kill points, and sequential milestones before Series B. Returns depend on successful pilot validation. This is not a guaranteed outcome.

16 TRANSACTION STRUCTURE & TERMS

Series A Investment Terms Summary

TERM	DETAIL
Investment Amount	$20M (full Phase 1 Series A)
Security	Ordinary Equity no preferences or special rights
Post-Money Valuation	$60.6M (33% equity at $20M raise)
Pre-Money Valuation	$40.6M
Investor Equity %	33% (fixed regardless of raise size between $17.5–20M)
Use of Funds	$16.2M deployed over 18 months (pilot, demo, team, BD); $3.8M strategic reserve / Series B bridge
Phase 1 Coverage	Full 18-month Phase 1 funded entirely from this raise no further dilution before Series B
Liquidation Preference	NONE pari passu with all shareholders on exit
Anti-Dilution Protection	Broad-based weighted average standard (Series B only)
Pro-Rata Rights	Investor has pro-rata participation right in Series B
Board Representation	Investor observer seat; monthly meetings (Months 0–18); then quarterly
Information Rights	Monthly P&L statements; quarterly board packs; annual audit
Exit Timeline	Strategic exit: 36–60 months post-pilot validation IP acquisition by Tier 1 industrial. Durability data (1,000+ hours), independent cost certification, and binding offtake agreement required before exit process.
Expected Investor Proceeds	Conservative: $52M · Base: $69M · Upside: $87M

NEXT STEPS FOR INTERESTED INVESTORS

WEEK 1

Week 1
Confidential Information Memorandum + signed NDA

WEEK 2

Week 2
Data room access (full financial model, technical reports, IP documentation)

WEEK 3

Week 3
Management presentation + Q&A session (virtual or in-person)

WEEK 4

Week 4
Reference calls: scientific advisors, potential customers, industry experts

WEEK 5–6	**Week 5–6** Term sheet negotiation and agreed principles
WEEK 7–8	**Week 7–8** Legal DD, subscription agreement, closing; funds received

▸ GigaCore is targeting a verified pathway to sub-$2.75/kg hydrogen. The Series A funds the full programme across 18 months with defined kill points. Investor returns depend on pilot validation and independent cost certification, not on financial projections or market assumptions.

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